

NIHARIKA VADLAMUDI

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EDUCATION	International Institute of Information Technology, Hyderabad (IIIT-H) [Part Time] M.S. by Research in Electronics & Communication Engineering Supervised by Prof. Ravi Kiran Sarvadevabhatla Thesis: High Precision Text Line Segmentation in Palm Leaf Manuscripts International Institute of Information Technology, Hyderabad (IIIT-H) B.Tech in Electronics & Communication Engineering	2022 - 2025 2018 - 2022
PUBLICATIONS	CARE-X: Towards Clinically Useful Radiology VLMs with Auxiliary Supervision, Reward-Aligned Learning, and Tool-Augmented Measurement Anirban Porya, Mercy Prasanna Ranjit, Niharika Vadlamudi*, and Nikhilesh Chowdary Eathamukkala, Prasanth V V, Sathvik Joel, Abhyuday Kumara Swamy, Pranay Narhari Umredkar, Pradeep Narayan, Vivek Rajagopal, Tanuja Ganu <i>Machine Learning for Healthcare (MLHC), 2026</i> Early Detection of Borderline Aortic Dilatation on Chest Radiographs Using a Multimodal Morphometric Agent Anirban Porya, Mercy Prasanna Ranjit, Niharika Vadlamudi, and Nikhilesh Chowdary Eathamukkala, Prasanth V V, Sathvik Joel, Abhyuday Kumara Swamy, Pranay Narhari Umredkar, Pradeep Narayan, Vivek Rajagopal, Tanuja Ganu <i>European Association for Cardio-Thoracic Surgery (EACTS), 2026</i> Rad-Phi4-Vision-CXR: A Compact Multimodal Assistant for Versatile Radiology Workflows Mercy Prasanna Ranjit, Anirban Porya, Shaury Srivastav, and Niharika Vadlamudi, Nikhilesh Chowdary Eathamukkala, Shashank Udyavar, Rahul Kumar, Tanuja Ganu <i>Machine Learning for Health (ML4H), 2025</i> LineTR: Unified Text Line Segmentation for Challenging Palm Leaf Manuscripts Vaibhav Agrawal, Niharika Vadlamudi, Muhammad Waseem, and Amal Joseph, Sreenya Chitluri, Ravi Kiran Sarvadevabhatla <i>International Conference on Pattern Recognition (ICPR), 2024</i> SeamFormer: High Precision Text Line Segmentation for Handwritten Documents Niharika Vadlamudi [†] , Rahul Krishna, and Ravi Kiran Sarvadevabhatla <i>International Conference on Document Analysis and Recognition (ICDAR), 2023</i>	
INDUSTRY EXPERIENCE	Microsoft Research <i>Research Intern, C.A.R.E Team</i> Advisor: Dr. Mercy Ranjit – Led a 5-person team bridging Narayana Health’s clinical needs and CARE-X model architecture for ICU chest X-ray pathology detection. – Coordinated clinical validation across 1,169 radiographs against exit criteria spanning 11 pathologies. – Trained the VQA model ranked 1st on Harvard’s RexRank (ReXVQA) leaderboard, beating 30+ models with about 94% accuracy. – Built a geometric, tool-augmented pipeline for ascending aortic enlargement detection pushing recall from 83.6% to 94.3% on 122 CT-confirmed cases — catching 40 of 43 mild dilation cases (93%) vs. 5 of 43 (12%) in initial radiology reads, surfacing 35 previously missed diagnoses. – Built an explainable Error Detection pipeline via GREEN / CRIMSON score flagging clinically significant report discrepancies. – Led tube/chest-line abnormality detection on a 5M-sample pipeline, fine-tuned via LoRA — 0.66 sensitivity / 0.77 specificity on Narayana Health’s ICU set.	Oct 2025 - June 2026

*First author – equal contribution.

[†]First author.

- Designed and trained auxiliary classification heads outperforming pure generative VQA — 0.943 sensitivity vs. 0.932 on ~6M curated dataset.

Microsoft

Aug 2022 - Aug 2025

Software Developer, Torus Buildout Team

- Built the first version of certificate provisioning in CredSMART — a configurable workflow adding support for publishing public trusted-root and public certificates to Azure AD, provisioning ~200 certificates every run.
- Ran integration testing across 30+ CredSMART APIs, supporting 100+ partner teams on secure key and certificate workflows.
- Automated core security services (Lockbox, IDM) via the automation tool, cutting manual effort by ~15%.
- Built the Torus Configuration Syncer Service, enabling dynamic, code-free config management across 5+ datacenter services.

Microsoft

Aug 2021 - Oct 2021

Intern, IC3 Admin Scenarios Team

- Enhanced pagination and search features in Modern Portal (MoPo) Admin, improving admin workflow efficiency and user experience.
- Designed and implemented Azure Cosmos DB (NoSQL) APIs to drive performance benchmarking and enable COGS optimization for custom Azure server configurations.
- Developed a scalable data ingestion tool handling millions of records, leading to 1M+ dollar cost savings across cloud operations.

IHub-Data

June 2022 - Aug 2022

Research Fellow, Molecular Intelligence Lab

Advisor: Dr. UD Priyakumar

- Worked in computational drug design on de novo molecular generation by extending SOTA baseline models for core property prediction on Harvard's TDC benchmark - ADMET (22 datasets, ~53K molecules).
- Designed GNN encoder-decoder architectures (GIN, GAT, GCN) for molecular property prediction, backed by literature review.

ACADEMIC PROJECTS

Context Encoders for Image Inpainting

- Implemented GAN model to learn the image context and reconstruct the missing image patch, with minimal loss of **4.5%** on Google's Street View Paris Building Dataset.

Multi-Source Approach for Crowd Density Estimation

- Designed a *Support Vector Regression (SVR)* model for crowd density estimation which uses multiple hybrid features, achieved minimal normalised loss of **0.46** on UCF Crowd Dataset.

Automatic Pathological Voice Detection

- Designed an automated Support Vector Machine model to extract vocal cord system features to classify normal human speech from voice disorder speech with a precision of **0.93**.

AWARDS

- **3rd Place, Microsoft Global Hackathon** – beat 20+ teams in the SovCloud organization 2024
- Awarded merit list for **top 30%** recognition for the Monsoon'21 semester at IIIT Hyderabad. 2021

RELEVANT COURSEWORK

Graduate: Mobile Robotics, Computer Vision, Machine Learning, Topics in Machine Learning, Deep Learning. *Undergraduate:* Robotics & Automation, Digital Image Processing, OOP & Data Structures, Operating System, Probability & Random Processes

SERVICES

- **NeurIPS 2021 Super Volunteer** – Led a team of 4 volunteers for 6+ hours at the Women in ML (WiML) 2021 help desk. 2021
- **Microsoft Diversity & Inclusion Lead** Oversaw 4 engineering managers and 1 group engineering manager on an 11-person D&I core team at Experiences + Devices. 2024
- **Open Source Contributor, IIT Kharagpur WoC-** Wrote 3,000+ lines of code across 5+ projects, mainly pothole-detection and a face-recognition attendance app. 2021